WESTBROOK

MATERIAL TEST REPORT

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Sold To: *MAVERICK INTERNATIONAL LTD* **Purchase Order:** *5466B* **Printed:** *5/31/2019*
Order No.: *792027* **Order Date:** *5/17/2019*

Line: **Description:**
20 *1 X 6 S80 A106B BLK SMLS NPL PXT*

****** Specifications - Heat Code W06323: ******
A106-18 GR.B, SA106 GR.B, A234-18 WPB, SA234 WPB, A530, A733, ASME B36.10, B1.20.1, MSS SP-25, SP-95
STARTING MATERIAL: A106B PIPE & TUBE

Chemical Composition %:

C:	Mn:	P:	S:	Si:	Cr:	Ni:	Mo:
0.186	*0.860*	*0.007*	*0.002*	*0.210*	*0.063*	*0.080*	*0.023*
Cu:	**V:**	**Al:**	**Cb/Nb:**	**Ti:**	**B:**	**C.E.:**	
0.210	*<0.001*	*0.003*	*0.002*	*0.002*	*.0001*	*.366*	

Mechanical Properties:

Tensile(PSI):	Yield(PSI):	Elong(%):	Hardness:	Bend:	Hydro(PSI):
74589	*50336*	*46.00*	*HRB 81-83*	*SATISFACTORY*	*2970*

Comments:
Normalized at 920C.
Hot finished.
Meets Nace MR-01-75/2015, MR-01-03/2012, ISO 15156/2015.

Line: **Description:**
21 *1/2 X 3 S160 A106B BLK SMLS NPL PXT*

****** Specifications - Heat Code W05368: ******
A106-15 GR.B, SA106 GR.B, A234-15 WPB, SA234 WPB, A530, A733, ASME B36.10, B1.20.1, MSS SP-25, SP-95
STARTING MATERIAL: A106B PIPE & TUBE

Chemical Composition %:

C:	Mn:	P:	S:	Si:	Cr:	Ni:	Mo:
0.190	*0.780*	*0.014*	*0.011*	*0.230*	*0.060*	*0.060*	*0.010*
Cu:	**V:**	**Cb/Nb:**	**Ti:**	**B:**	**C.E.:**		
0.150	*0.007*	*0.001*	*0.001*	*<.0001*	*.349*		

Mechanical Properties:

Tensile(PSI):	Yield(PSI):	Elong(%):	Hardness:	Bend:	Hydro(PSI):
72650	*50320*	*38.20*	*<= HRC 22*	*GOOD*	*2970*

Comments:
Normalized.
Cold drawn.
Meets Nace MR-01-75/2015, MR-01-03/2012, and ISO 15156/2015.

****** Specifications - Heat Code W05370: ******
A106-15 GR.B, SA106 GR.B, A234-15 WPB, SA234 WPB, A530, A733, ASME B36.10, B1.20.1, MSS SP-25, SP-95
STARTING MATERIAL: A106B PIPE & TUBE

Chemical Composition %:

C:	Mn:	P:	S:	Si:	Cr:	Ni:	Mo:
0.190	*0.810*	*0.013*	*0.009*	*0.230*	*0.050*	*0.050*	*0.020*
Cu:	**V:**	**Cb/Nb:**	**Ti:**	**B:**	**C.E.:**		
0.140	*0.006*	*0.001*	*0.001*	*.0003*	*.352*		

We hereby certify that the contents of this Material Test Report are correct and accurate to the best of our knowledge and belief, that all test results are in compliance with the material specifications and that all material meets the dimensional requirements of the order. The products described have not come into contact with mercury or any of its components. No weld repair performed. Westbrook's Quality System is registered to ISO 9001, Certificate #32813. This document is in accordance with EN10204:2004 3.1/DIN50049 3.1B and was electronically transmitted. This inspection certificate is issued under the authority of Westbrook's QA Manager.

Prop 65 Warning for California Residents - Products listed may contain chemicals including Nickle and/or Carbon known to the State of California to cause cancer, birth defects and other reproductive harm. http://www.p65warnings.ca.gov/